

FORGE — FINE AUCTIONS

Real-Time Bidding × Angular 21

Angular 21

TypeScript 5.x

WebSocket STOMP

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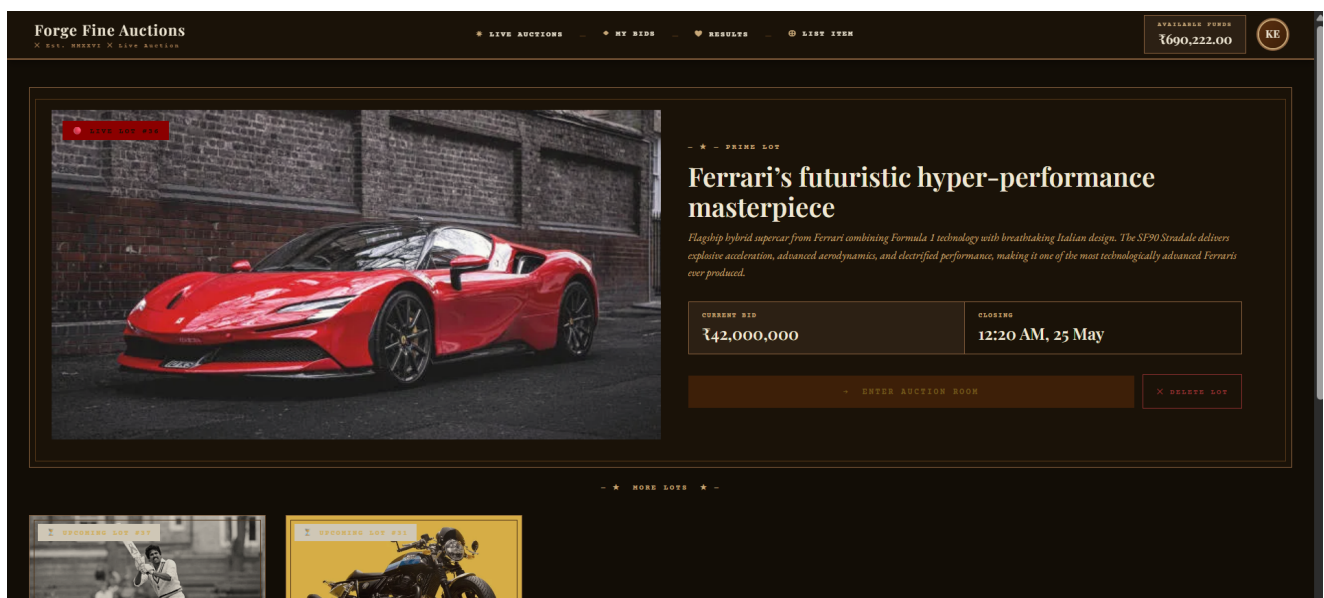
A vintage-aesthetic, real-time auction frontend — built to feel like a Christie's catalogue, engineered to move like a trading floor.

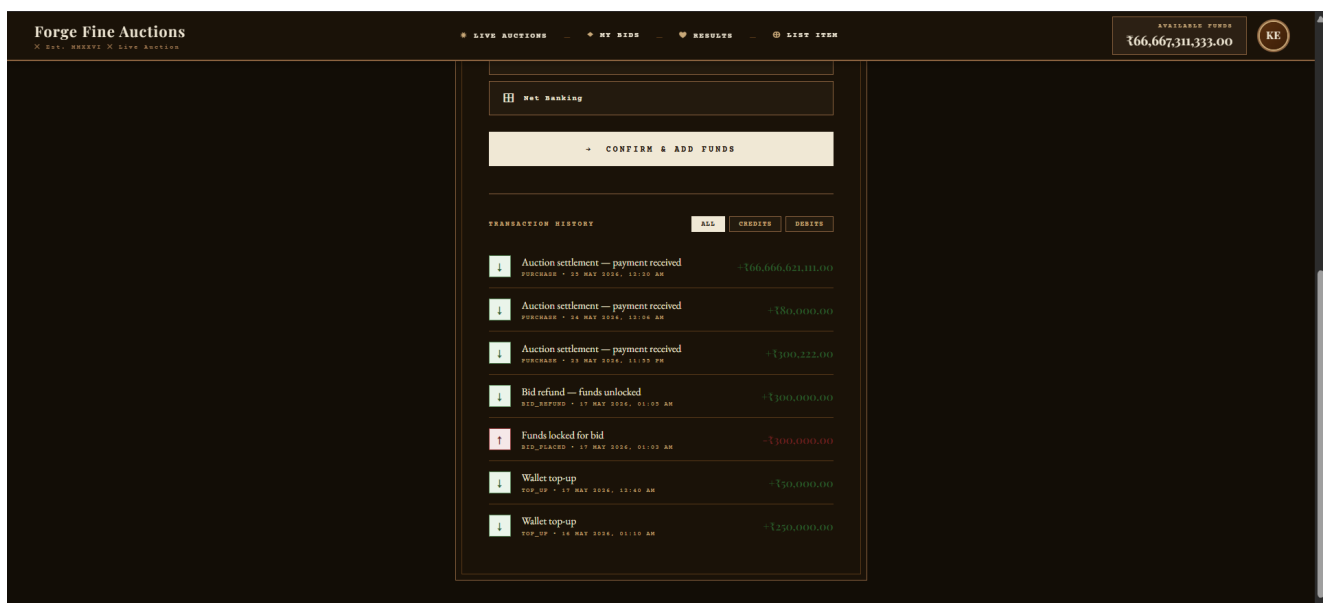
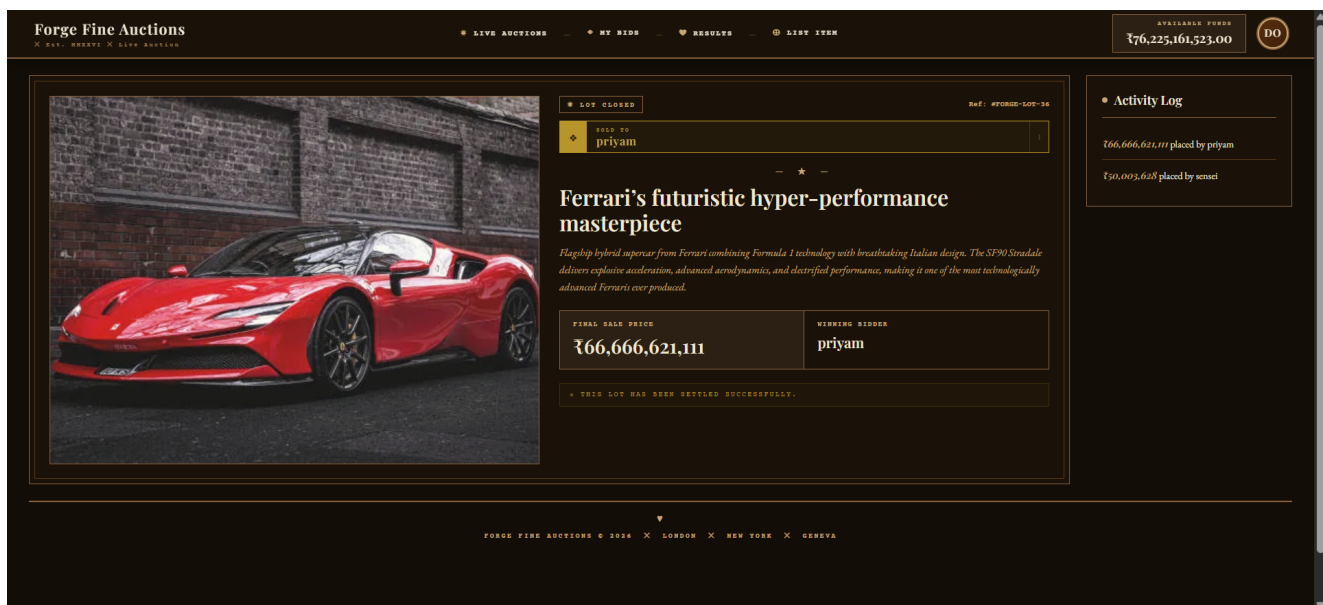
[Backend →](#)

Overview

`forge-engine-ui` is the Angular frontend for the **FORGE auction platform** — a production-grade, real-time bidding system. While most auction UIs feel transactional and cold, FORGE leans into the theatre of live bidding — parchment tones, serif typography, countdown clocks, and live WebSocket feeds that update without a single page refresh.

This is not a CRUD app with a table. This is a bidding room.





Stack

Layer	Technology
Framework	Angular 21 (Standalone Components)
Language	TypeScript 5.x
Real-time	WebSocket + STOMP over SockJS
Styling	Pure CSS (custom properties, no UI library)
HTTP	Angular HttpClient
Routing	Angular Router with lazy loading
Auth	JWT — stored in localStorage

Features

LIVE AUCTIONS	Real-time bid updates via WS
COUNTDOWN TIMER	Live end-time clock per auction
ACTIVITY LOG	Streaming bid history, live
WALLET	Top-up, balance, tx history
MY BIDS	Active bids + won/lost status
LIST ITEM	Create & schedule new auctions
RESULTS	Completed auction outcomes

- **Real-time bidding** — WebSocket/STOMP connection to forge-engine; bid placed → all connected clients update instantly
- **Live countdown** — per-auction timer, auto-extends on last-minute bids (anti-sniping)
- **Wallet system** — top-up funds, locked/available balance, paginated transaction history
- **JWT auth** — login/register flow, token-gated routes
- **Parchment aesthetic** — dark warm palette, Playfair Display headings, vintage serif throughout
- **Responsive** — works across desktop and tablet viewports

Project Structure

```
src/app/
├── components/
│   ├── auction-lobby/      # Live auctions grid + featured lot
│   ├── auction-detail/    # Bidding room – WS, timer, activity log
│   ├── create-auction/    # List a new lot
│   ├── wallet/            # Funds management
│   ├── my-bids/           # User's active bids
│   ├── results/           # Completed auctions
│   ├── profile/           # User profile
│   ├── login/             # Auth
│   ├── register/         # Auth
│   └── navbar/            # Global navigation
├── services/
│   ├── wallet.service.ts  # Balance + transaction API
│   └── websocket.service.ts # STOMP connection management
└── guards/               # Route auth guards
```

Getting Started

Prerequisites: Node 18+, Angular CLI 21, FORGE backend running on :8080

```
# Clone
git clone https://github.com/Ketansinghrajput/forg-engine-ui.git
cd forge-engine-ui

# Install
npm install

# Run (dev server)
ng serve

# Open
# http://localhost:4200
```

Backend required — start **FORGE** first (Docker Compose recommended):

```
# In the FORGE repo
docker compose up -d
```

WebSocket Architecture



On **auction-detail** load:

1. JWT fetched from localStorage
2. STOMP connection established with auth header
3. Subscribed to auction topic + personal error queue
4. Bid placed → optimistic UI update → confirmed/rejected via WS

UI Highlights

Auction Lobby — featured hero lot + paginated grid. Status-aware cards: **ACTIVE** opens bidding room, **PLANNED** opens detail/preview.

Bidding Room — live price tracker, countdown timer with anti-snipe extension indicator, activity log streaming in real-time, available funds validation before bid submit.

Wallet — preset amounts + custom input, mock payment flow (UPI/Card/Net Banking), paginated transaction history with CREDIT/DEBIT filter.

Built by [Ketansingh Rajput](#)

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